

Total No. of Questions : 10]

SEAT No. :

P3623

[Total No. of Pages : 3

[5154]-534

B.E. (Mechanical) (Semester - I)
ENERGY AUDIT AND MANAGEMENT
(2012 Pattern) (Elective - I)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Neat diagram must be drawn wherever necessary.*
- 2) *Figures to the right indicates full marks.*
- 3) *Use of Electronic pocket calculator is allowed.*
- 4) *Assume suitable data, if necessary.*

Q1) a) Write a short note on:

[6]

- i) Commercial & non-commercial energy.
- ii) Renewable & non-renewable energy

b) Explain preliminary energy audit & its importance.

[4]

OR

Q2) a) Explain detailed energy audit with its 10 steps.

[6]

b) Explain following instruments used in Energy Audit with their application:

[4]

- i) Combustion analyzer
- ii) Infrared pyrometer

P.T.O.

- Q3) a)** What is the NPV of an energy conservation project with cash flow given below: [6]

Initial investment	Rs. (1,000,000)
Saving in Year	Cash Flow
1	Rs. 200,000
2	Rs. 200,000
3	Rs. 300,000
4	Rs.300,000
5	Rs. 350,000

The discount rate $k = 10\%$. Is the proposal attractive?

- b) Write a short note on Return on Investment. [4]

OR

- Q4) a)** An economizer costs Rs. 200000 and will last for 10 years. It will generate a saving of Rs. 35,000 per year with a maintenance cost of Rs.5000 per year. The discount rate is 10% and salvage value is Rs.5000 at the end of 10 years. Is the proposal attractive by NPV method? [8]

- b) What is the simple payback period? [2]

- Q5) a)** Explain merits and demerits of Direct and Indirect methods used for calculating boiler efficiency. [8]

- b) Write down various energy conservation opportunities in HVAC System and DG set. [8]

OR

- Q6) a)** Explain various losses in industrial furnace system with a neat schematic sketch. Explain the indirect method of performance evaluation of a furnace. [8]

- b) What is the importance of Analysis and Recommendation for Energy Audit? [8]

- Q7)** a) Explain electrical billing tariff structure for an industrial consumer. [8]
b) What is power factor? What are the benefits of improving power factor? [8]

OR

- Q8)** a) What is maximum demand in electrical systems? Explain how maximum demand is calculated with the help of a load curve. [8]
b) What are the types of lamps used in lighting system? Write down their features with typical applications. [8]

- Q9)** a) Describe various factors influencing selection of co-generation plant. [8]
b) Write short note on: [6]
i) CDM projects
ii) Carbon credits
c) How the Waste heat recovery systems are classified? [4]

OR

- Q10)** a) What is cogeneration? What are the advantages of cogeneration? [8]
b) What are the direct and indirect benefits of Waste Heat Recovery plant? [6]
c) Explain working of convective Recuperator with neat sketch. [4]
